Dial Gauge



Category: **Measuring & Marking Tools** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Dial gauge

What it is used for

To measure small linear movement, runout, or dimensional variation using a dial gauge. To obtain repeatable readings with correct alignment and a stable setup.

Main parts

- Dial face and pointer
- Graduated scale
- Contact plunger or probe
- Mounting stem or clamp
- Magnetic base or fixture, if supplied

Before you start

1. Inspect the dial face, probe, and mounting parts for damage.
2. Confirm the gauge range and resolution suit the measurement.
3. Clean the gauge contact point and workpiece surface.
4. Secure the gauge stand and workpiece firmly.
5. Check zero setting and calibration status where required.

How to use

1. Mount the dial gauge securely on a stable stand.
2. Position the probe against the measurement surface at the recommended angle.
3. Apply the specified preload without exceeding the gauge travel.
4. Set the dial to zero or record the initial reading.
5. Move or rotate the workpiece slowly as required.
6. Observe and record the pointer movement or maximum variation.
7. Remove the gauge carefully and compare results with the required tolerance.

⚠ Safety

- Keep fingers clear of rotating or moving machinery.
- Never exceed the probe travel limit.
- Do not force the probe sideways.
- Stop machinery and isolate it before mounting or adjusting the gauge when required.

Care & storage

- Wipe the probe and gauge body clean after use.
- Retract and protect the probe during storage.
- Avoid shocks, drops, and moisture.
- Arrange calibration checks according to workplace requirements.

Common mistakes to avoid

- Mounting the gauge on an unstable base.
- Misaligning the probe with the measurement direction.
- Exceeding the gauge travel range.
- Forgetting to zero the dial or record the initial reading.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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